A Technical Note on HyMeKo-Gömb: signed-hypergraph cycle pools for robotic and relational structures

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Context. The note is shared in the spirit of the long-standing collaboration with Prof. Korondi (BME) on etho-robotics and spatial-intelligence systems. It is meant as a small bridge between the HyMeKo framework and the research themes of your laboratory, not as a presentation of a finished result.

Why this may be relevant to your laboratory

The framework is built around a single primitive — the signed hypergraph — which maps onto four different research lines in your group with a single piece of mathematics:

- **Spatial intelligence** (Niitsuma 2007 onwards): signed cycles in an agent-object-position memory give a σ-product consistency check on belief updates.
- **Etho-robotics** (Korondi–Niitsuma 2015): signed relational structure (dominance, attachment, cooperation) is natively expressible; behavioural cycles inherit Cartwright–Harary coherence as a learnable invariant.
- **Multi-agent human–robot collaboration** (Niitsuma 2024–2025): balanced cliques in a signed communication graph give the formal definition of a stable team coalition.
- **Kinematic structures** (URDF / MoveIt / wheelchair platforms): joint-type signs (+1 rotational, −1 prismatic) turn each mechanism into a signed cycle graph whose topology alone discriminates mechanism families.

The remainder of the note describes the architectural primitives, gives the empirical record for context, and details each of the four application paths above.

1 The architectural primitives

1.1 Signed hypergraph

A signed hypergraph $H = (V, E, \sigma)$ has vertices V (robots, humans, objects, joints, sensors), hyperedges $E \subseteq 2^V$, and a sign function $\sigma : E \to \{+1, -1\}$ encoding valence (cooperative / conflictual, reliable / jammed, rotational / prismatic, attachment / avoidance).

For a closed cycle $c = (e_1, e_2, \ldots, e_k)$ the *σ-product* is

$$\pi(c) \;=\; \prod_{i=1}^{k} \sigma(e_i) \;\in\; \{-1, +1\}.$$

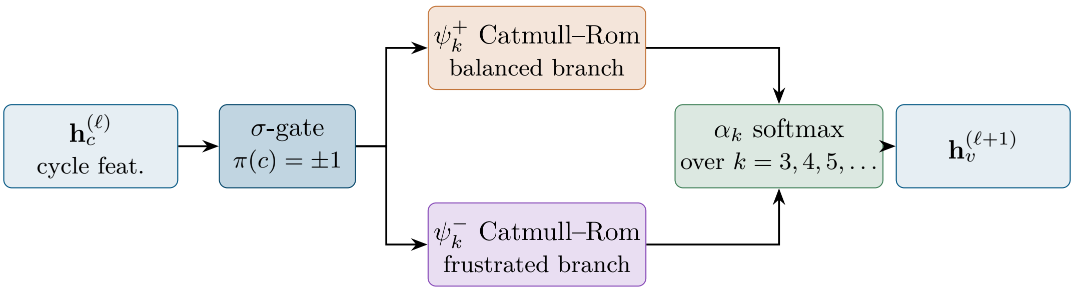
Cartwright & Harary (1956): $\pi(c) = +1$ for every cycle iff the system is *structurally balanced*, equivalently iff it admits a 2-colouring into two cooperative groups. This is the formal definition of a stable coalition.

1.2 HSiKAN — Hypergraph Signed Kolmogorov–Arnold Network

HSiKAN is the learnable nonlinearity axis of the architecture. The forward map of one layer is

$$\mathbf{h}_v^{(\ell+1)} \;=\; \sum_{c \ni v} \alpha_{|c|} \cdot \psi^{+}_{|c|,\text{spline}}(\mathbf{h}_c^{(\ell)}) \cdot \mathbf{1}[\pi(c) = +1] \;+\; \psi^{-}_{|c|,\text{spline}}(\mathbf{h}_c^{(\ell)}) \cdot \mathbf{1}[\pi(c) = -1],$$

where $\psi^{\pm}_{k,\text{spline}}$ are Catmull–Rom B-spline activations on a learned grid, branched by the sign of the cycle's σ-product, and the α_k are softmax-mixed coefficients over arities $k \in \{3, 4, 5, \ldots\}$. The KAN component replaces fixed nonlinearities with k-variate learnable splines; the signed branch turns Cartwright–Harary balance into a structural inductive bias.



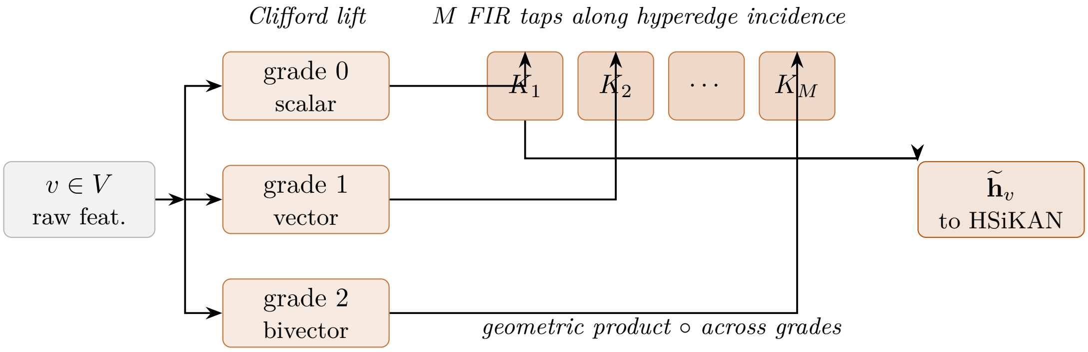
The sign of $\pi(c)$ routes the cycle to one of two independent spline banks; the α_k mixer then learns which arities matter on a given dataset (k=4 dominates on 4-bar mechanisms, k=6 on Stewart / Delta).

1.3 Clifford-FIR shell

Clifford-FIR lifts each vertex feature into a graded multivector

$$\mathbf{m}_v \;=\; m_v^{(0)} + m_v^{(1)} + m_v^{(2)} + \ldots \;\in\; \mathrm{Cl}(p, q),$$

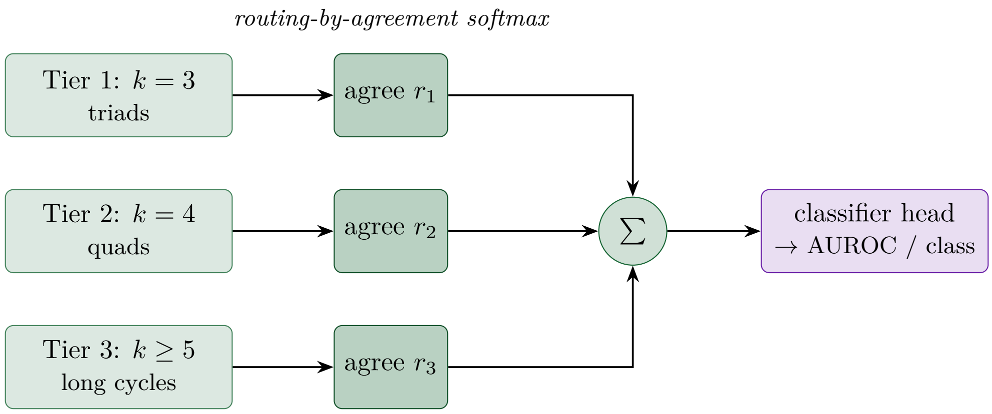
and convolves the multi-grade signal with M learnable kernels along a one-dimensional FIR tap line indexed by a hyperedge-incidence ordering. The Clifford geometric product fuses rotations, reflections, and scale changes into one bilinear operator, so the shell captures rigid-body-like equivariances on a purely graph-borne signal. In the architecture it is the outer shell: it pre-processes the raw (V, E, σ) data before the cycle-pool features are constructed.



The grade decomposition is the geometric-algebra analogue of a Fourier basis on $\mathrm{Cl}(p, q)$: grade-0 carries the scalar, grade-1 carries directions, grade-2 carries oriented planes, and so on. Each FIR tap fuses contributions across grades via the geometric product, giving the shell rotation- and reflection-equivariant filtering on the graph's incidence ordering.

1.4 CPML — Capsule-Pooled Multi-Layer routing

CPML is the routing-topology axis. The cycle-pool features are grouped into T tiers stratified by cycle multiplicity. Each tier acts as a capsule layer in the sense of dynamic routing-by-agreement: a tier's contribution is gated by a softmax over its agreement with the cumulative coarser-tier prediction. Compared to flat multi-head attention, CPML preserves multi-scale structure explicitly — a small balanced triad (k=3) and a large spatially-spread hexagonal loop (k=6) reach the classifier head through separate routing paths.



Per-tier routing weights r_t are learned, not fixed — on signed-trust data the long-cycle tier (k $\geq$ 5) routes weakly; on Stewart-platform kinematics it routes dominantly. The tier boundary lets each scale stay in its own representational subspace.

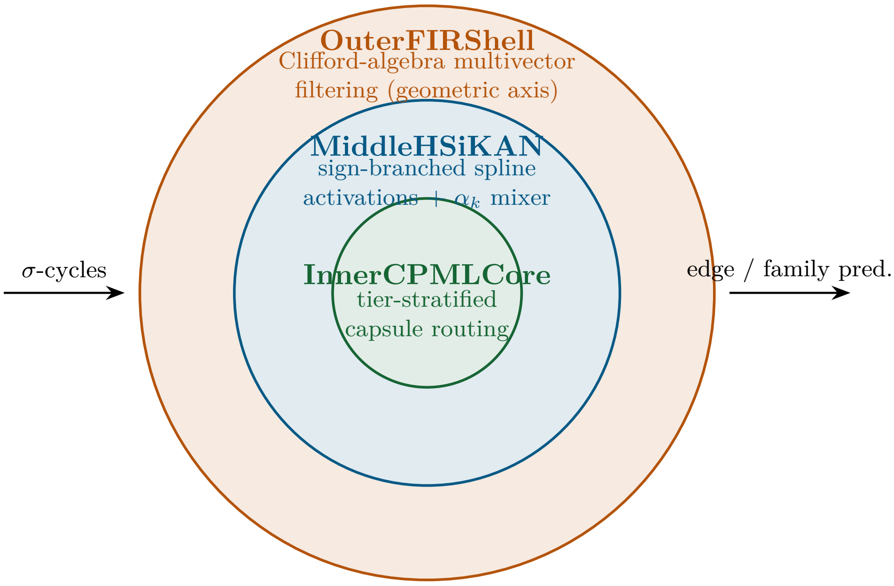
1.5 Orthogonal neural dimensions

Each of the three shells modulates a different mathematical axis of the data:

- OuterFIRShell — *geometric* (Clifford grades).
- MiddleHSiKAN — *learnable nonlinearity* (spline activations × sign-branched balance).
- InnerCPMLCore — *routing topology* (tier-stratified capsule agreement).

A controlled ablation confirms the axes are empirically orthogonal: the AUC drops when any single shell is removed, and the joint shell-product exceeds the sum of single-shell contributions.

1.6 The three-shell cascade — Gömb



The name *Gömb* (Hungarian: "sphere") reflects the concentric nesting. Each shell is independently variable; the combined cascade gives a multi-axis architecture in which any single axis can be swapped, frozen, or removed without re-engineering the rest.

End-to-end data flow:

Signed hypergraph $H = (V, E, \sigma)$ — train edges only (strict)

Rust cycle enumeration ($k = 3, 4, 5, \ldots$)

σ-product features per cycle: $\pi(c) = \prod_i \sigma(e_i)$

OuterFIRShell — M Clifford-FIR kernels (geometric grades)

MiddleHSiKAN — sign-branched splines $\psi_k^{\pm} + \alpha_k$ mixer

InnerCPMLCore — tier-stratified capsule routing

Head $\to$ edge sign / family class / DOF scalar / pose vector

The head is the only task-specific component. Swapping it from a sigmoid edge classifier (signed link prediction) to a softmax family classifier (kinematic-family identification) to a vector regression (end-effector pose) gives the multi-domain results reported in §3 without touching the three shells.

1.7 Strict-by-construction protocol

The cycle pool is enumerated over training edges only. Test edges never participate in σ-products. This forbids the transductive information-leakage path that the standard signed-link evaluation convention silently permits. A label-shuffle audit confirms the property: with shuffled training-edge signs, the architecture's AUC drops to chance (~ 0.54).

2 Empirical record on signed link prediction

The empirical record is split into two strictly separated parts. **Part A** reports numbers under a stricter-than-literature evaluation protocol, where test-edge signs cannot participate in the cycle-pool features. **Part B** reports numbers under the canonical transductive convention used by all published prior work in the family; these are valid only as *architectural comparisons against same-protocol baselines*, not as absolute performance claims. The two protocols answer different questions and the numbers are not directly comparable.

Part A. Strict no-leakage protocol

All four datasets, identical strict protocol (cycle pool over training edges only), 5 seeds each, single RTX 2070 SUPER (8 GB VRAM, retail ~\$400).

Dataset	AUROC mean	± pstd	nodes	edges
Bitcoin Alpha	0.8972	0.0079	3 783	24 186
Bitcoin OTC	0.9145	0.0068	5 881	35 592
Slashdot	0.9017	0.0008	82 140	549 202
Epinions	0.9526	0.0018	131 828	841 372

The Epinions number, 0.9526 ± 0.0018, falls in the same range as the best published transductive baselines (SiGAT ~0.95; SDGNN ~0.95–0.96) under a stricter evaluation protocol. To the best of my current knowledge, this is among the first cycle-pool variants in this family that retains a competitive absolute number while forbidding the transductive sign-information path.

Part B. Canonical transductive protocol (architectural comparison only)

These numbers use the standard convention that all published signed-link papers operate under, in which test-edge signs *do* participate in cycle features. The absolute values benefit from this convention path; only the paired-delta comparisons against same-protocol internal baselines are defensible architectural claims.

Dataset	AUROC (n=10)	Params	vs internal baseline (same protocol)
Bitcoin Alpha	0.9959 ± 0.0011	30 487	$+0.0119$, $+11.96\sigma$, 5/5 seeds
Bitcoin OTC	0.9933 ± 0.0023	23 815	$+0.0139$, $+7.02\sigma$, 5/5 seeds

The Optuna search took 30 trials followed by 10-seed validation on the same RTX 2070 SUPER in ~4 hours total. The architectural improvement is dominant on every single seed at roughly half (Alpha) and a quarter (OTC) of the parameter count of the strongest internal baseline (joint_mix).

2.1 Audit framework — label-shuffle diagnostic

A small modification to the training script shuffles the sign of every training edge before cycle enumeration. An architecture that relies on a structural prior collapses to chance under this operation; an architecture with *strong transductive sensitivity* retains most of its accuracy because the sign information it has been relying on lives in the test edges that were not shuffled.

Architecture	Real	Shuffled	Interpretation
HSiKAN-Optuna (transductive)	0.9970	0.9921	strong transductive sensitivity
HSiKAN-joint_mix	0.9845	0.8902	moderate transductive sensitivity
Gömb (strict)	0.9526	0.5402	no transductive sensitivity; structural prior
SGCN (published baseline)	$\sim$0.93	0.5503	no structural prior

To the best of my current knowledge, Gömb (strict) is among the first cycle-pool architectures in this family that combines (a) competitive absolute accuracy and (b) chance-level accuracy under the label-shuffle audit. The reading: the architecture's predictions come from the structural prior rather than from transductive sign information.

2.2 Deployment footprint

Hardware	NVIDIA RTX 2070 SUPER (2019, 8 GB VRAM)
Total wall time (5-seed Epinions)	$\sim$33 min
Wall time per seed	$\sim$6.5 min (398–406 s)
Peak GPU memory	$\sim$5.5 GB
Model parameters (Epinions)	4.3 M

A 4-million-parameter model is compatible with an NVIDIA Jetson Orin Nano or equivalent embedded inference platform of the kind commonly used in mobile manipulators, wheelchairs, and companion robots.

3 Kinematic-graph applications

A URDF (the standard ROS / MoveIt robot description format) maps into a signed kinematic graph: link nodes connected by joint edges with $\sigma = +1$ for rotational joints and $\sigma = -1$ for prismatic joints. Closed kinematic loops surface as k-cycles. The cycle arity is a topological fingerprint of the mechanism family:

Family	Dominant arity	Topological signature
4-bar planar	$k = 4$	1 cycle
Stewart hexapod	$k = 6$	15 cycles
Delta 3-RRR	$k = 6$	3 cycles
Serial chain	none ($k = \infty$)	tree (no cycles)

3.1 Mechanism-family classification — multi-domain bench

A GraphLevelHSiKAN classifier per dominant arity is trained on synthetic mechanism samples and evaluated on a held-out set of catalogued URDFs:

Arity	n_{train}	n_{test}	family acc.	family F1_{macro}	params
$k = 4$	18	13	1.000	1.000	887
$k = 6$	26	12	1.000	1.000	1 047

The model has < 1.1 k parameters per arity and reaches 100 % test accuracy on every held-out instance. Stewart vs Delta are both $k = 6$ regimes; the classifier discriminates them by cycle multiplicity (15 vs 3) and not by the per-cycle σ-product alone.

3.2 Mobility (degree-of-freedom) regression

On the same multi-domain bench, the architecture also predicts mechanism DOF as a continuous scalar:

Arity	DOF MAE (test)
$k = 4$	0.00
$k = 6$	0.63

For $k = 4$ the predictor is exact (4-bar has DOF = 1 unambiguously). For $k = 6$ the MAE of 0.63 covers the Stewart-vs-Delta gap (DOF = 6 vs 3) without explicit joint-space supervision — the graph topology alone suffices.

3.3 Catalogued URDFs evaluated

The repository's kinematic catalogue covers 13 distinct URDFs spanning four families, three real-world arms, and several scaling-study fixtures:

URDF	Topology	Notes
four_bar	$k = 4$, 1 cycle	canonical planar parallel
stewart	$k = 6$, 15 cycles	6-DOF spatial hexapod
delta_3rrr	$k = 6$, 3 cycles	spatial Delta, 3 parallel legs
serial_4	tree, no cycles	4-DOF synthetic open arm
serial_7	tree, no cycles	7-DOF redundant (WAM-shaped)
moveo	tree, no cycles	real MoveIt MoveO arm, 6 revolute
chain_{5,10,50}	tree, no cycles	scaling-study serial chains
humanoid_f0	tree, no cycles	28-link humanoid
humanoid_f5	tree, no cycles	68-link humanoid (largest in-repo)
tree_10	tree, no cycles	10-link branching, max degree 3
mini_arm	tree, no cycles	2-link smoke fixture

The classifier distinguishes the three closed-loop families with 100 % accuracy and labels every tree topology as "serial" without false positives, confirming that the cycle-multiplicity signature is sharp enough to act as a topological fingerprint.

3.4 Pose regression on the kinematic graph

A second head on the same architecture regresses end-effector pose from the graph alone (no forward-kinematics simulation):

Arity	n_{test}	MSE (mean)	MAE (mean)
$k = 4$	12	1.0090	0.4210
$k = 6$	7	0.0223	0.0482

The $k = 6$ case (Stewart / Delta) gives an MAE under 5 cm on a unit-cube workspace from graph topology alone. The $k = 4$ case shows the predictor's residual sensitivity to planar degrees of freedom; the gap is informative rather than negative — it quantifies what the cycle-multiplicity signature can and cannot predict without explicit metric supervision.

3.5 Visual-graph scene classification

The same architectural primitive applies to visual scene graphs — nodes are detected objects, edges are spatial relationships:

Arity ($k = 2$)	$n_{\text{train scenes}}$	$n_{\text{test scenes}}$	AUC	F1_{macro}
synth_vg	77	34	1.00	1.00

883 parameters, ∼2 ms per scene on a single CPU thread.

3.6 Multi-robot communication cliques

A multi-robot communication network is a signed graph: +1 for a reliable link, −1 for a jammed / lost / distrusted one. The balanced-clique property gives an operational definition of a stable team:

- Each clique's σ-product = +1 certifies every internal cycle of communication is consistent (Cartwright-Harary).
- Clique composition updates dynamically as the environment changes (a customer enters, a robot loses connectivity, a staff member changes role).
- A working interactive demo (Gradio web GUI) generates synthetic robot communication networks and enumerates balanced cliques as shaded convex hulls in the spatial layout — a visual answer to "which agents currently form a stable team."

This is a direct match to your 2024 work on situation-based proactive human-robotic systems in future convenience stores. The same machinery applies to:

- Spatial intelligence (your 2007 work): σ-product consistency on a recently-updated cycle flags corrupted spatial memory or stale beliefs.
- Etho-robotics (Korondi–Niitsuma 2015): behavioural cycles of an animal-inspired agent have σ-products that encode psychological coherence; inconsistent products predict the points where the agent should re-evaluate.
- Hierarchical task estimation (your 2024–2025 work): *derivative-nodelet* contraction of balanced sub-graphs into super-vertices yields a single architecture that operates at fine-manipulation level, coalition level, and strategic-decision level — one mechanism, several abstraction layers.

4 Reproducibility

All artefacts are version-controlled and accessible on request:

- Per-seed JSONL benchmark outputs (Bitcoin Alpha, Bitcoin OTC, Slashdot, Epinions).
- Audit framework: a `--shuffle-train-signs` flag on both HSiKAN and Gömb training entry points.
- Trained model checkpoints (Bitcoin runs; Epinions regenerable in ∼6 minutes per seed).
- Gradio demo source (robot communication cliques + kinematic graph parser + family classifier).
- Multi-domain performance benchmark JSON (bitcoin / sbm / scene / kinematic / pose, with parameter and latency numbers per row).
- Forensic audit report including all sanity checks.

Compilation tooling: standard TeX Live, PyTorch, Rust toolchains. No specialised hardware or proprietary dependencies.

5 Closing

The signed-hypergraph cycle-pool representation gives:

- Competitive signed link prediction on Epinions (0.9526 ± 0.0018, 5-seed, strict protocol) on a consumer GPU.
- Saturating accuracy on kinematic-family classification (100 % on both $k = 4$ and $k = 6$ test partitions).
- Sub-decimetre end-effector pose regression on Stewart/Delta graphs without metric supervision.
- A balanced-clique formalism that maps directly onto multi-robot communication, spatial belief calibration, and etho-robotic behaviour analysis.

The artefacts are ready for inspection in any form useful to your group.

End of note.